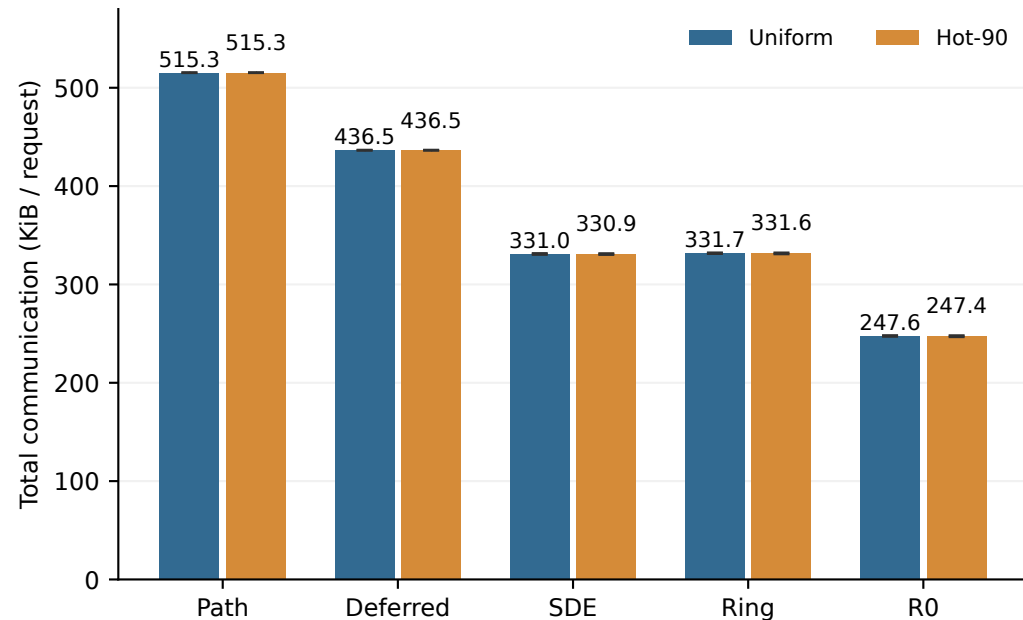
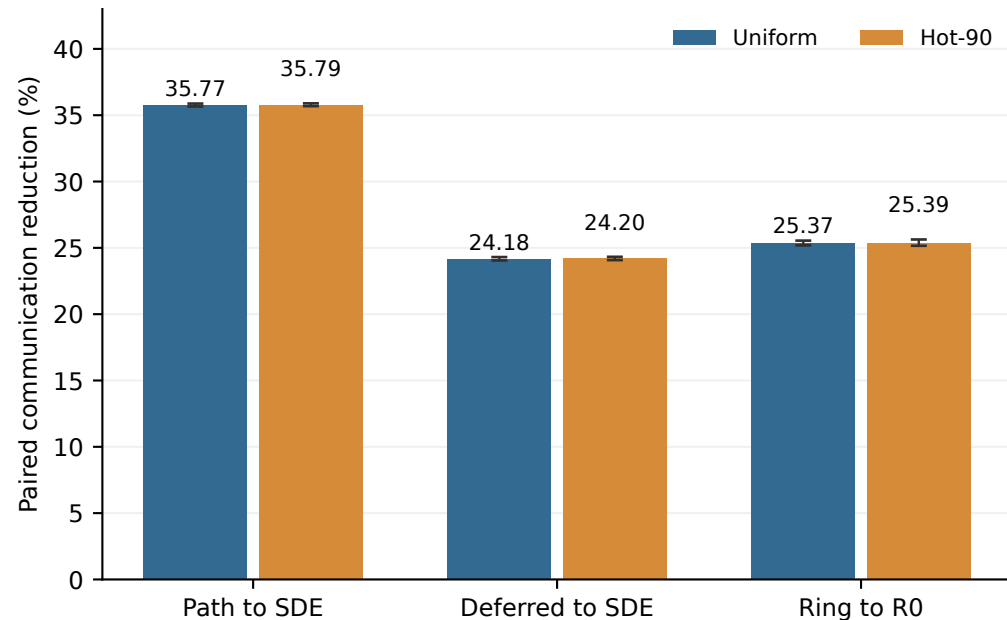


(a) Absolute cost after separate tuning



(b) Reduction vs each stated baseline



N = 16,384; data B = 4,096 bytes; map B = 256 bytes; R = 256; five paired seeds, mean and 95% t intervals.
Path / Deferred / SDE: Z4/A3; Ring: Z7/A6/S9; R0: Z7/A6/S6. 2,048 warmup + 4,096 measured requests.

All serialized protocol bytes, including authentication and recursive maps. Predeclared search: 72 candidates.
Common resource ceilings; actual storage differs. Finite window, empirical Path Z4, no global-optimum or latency claim.